

REVIEW ARTICLE



Research progress and development strategies of antibody-oligonucleotide conjugates

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Antibody-oligonucleotide conjugates (AOCs) effectively integrate the delivery capability of antibodies with the specific gene regulatory function of oligonucleotides, offering a novel strategy for extrahepatic delivery. Unlike antibody-drug conjugates (ADCs), AOCs employ nucleic acid payloads, enabling specific gene modulation with reduced off-target effects, thus providing new avenues for treating genetic disorders. While AOC candidates for conditions such as Duchenne muscular dystrophy (DMD) have progressed to Phase III trials, their development remains constrained by limited targets like Tfr1, and their activity is highly dependent on antibody selection, linker design, and modification strategies. This study delves into the fundamental principles of target selection and antibody engineering, emphasizing that the endocytic efficiency of the target receptor is a critical factor for the successful delivery of AOC therapeutics. Taking Tfr1 as an example, it analyzes the advantages and disadvantages of different antibody formats, examines the structure-activity relationships between linker chemistry and pharmacokinetics/pharmacodynamics, and further explores nucleic acid modification strategies aimed at enhancing delivery efficiency. Finally, the study outlines future directions for AOC development, including advances in bispecific antibodies, peptide conjugation, gene editing, and artificial intelligence-driven approaches, aiming to provide forward-looking perspectives and a theoretical foundation for the design of AOC therapeutics.

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INTRODUCTION

The development of tiny nucleic acid therapies is largely fueled by advances in ligand-conjugated delivery technologies. By focusing on the asialoglycoprotein receptor (ASGPR), the N-acetylgalactosamine conjugation platform (GalNac) achieves extremely effective and targeted transport of small interfering RNA (siRNA) to the liver. RNA interference therapy has been effectively extended from rare diseases to major chronic disease areas with the approval of seven GalNac-siRNA medicines for market release [1–3]. Benefit from GalNac, Inclisiran (Novartis) allows for twice-yearly dosing, which is a crucial advantage for the long-term treatment of chronic illnesses such as amyloidosis, hypercholesterolemia, and several genetic muscle disorders [4, 5]. The remarkable liver targeting of the GalNac platform stems from a synergy between active targeting via the hepatocyte-specific ASGPR and the liver's inherent propensity for passive uptake due to its high blood flow and sinusoidal endothelium. In contrast, extrahepatic delivery remains a significant challenge [6, 7].

To overcome the bottleneck of extrahepatic delivery, antibody-oligonucleotide conjugates (AOCs) have emerged as an innovative strategy. By employing chemical linkers, AOCs combine the specific targeting capability of antibodies with the gene-regulating function of oligonucleotides. By precisely delivering

oligonucleotides to particular cell types, this integration seeks to degrade or functionally regulate target mRNAs [8]. This idea originated in 2005 when Judy Lieberman's group at Harvard University used protamine to noncovalently bind siRNA that targeted the HIV-1 capsid protein to antibody Fab fragments. This method provided proof-of-concept validation by successfully suppressing viral expression in infected T cells [8, 9]. In 2015, a study led by researchers at Genentech successfully achieved the site-specific and covalent conjugation of siRNA to antibodies by the first time to use the company's proprietary Thiomab™ platform (a site-specific conjugation technology) [10]. This combination may be successfully targeted and delivered to tumor tissues, resulting in selective gene silencing, according to preclinical validation in mice tumor models supplying crucial experimental support for the AOC theory. However, AOC molecules still face significant drawbacks because unmodified siRNA is prone to degradation in vivo, leading to poor stability, reduced duration of effectiveness, and unsatisfactory therapeutic outcomes. Concurrently, established nucleic acid chemical modification techniques used in GalNac delivery platforms can effectively suppress innate immune activation triggered by immunostimulatory siRNAs while enhancing their activity and specificity. These modifications include substituting the 2'-hydroxy

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Table 1. AOC conjugates on clinical stage.

Drug	Company	Target	Linker	Antibody	Oligonucleotide	Indications	Phase
AOC-1001	Avidity	DMPK	SMCC	TfR1-targeted mAb	DMPK-siRNA	DM1	Phase III
AOC-1020	Avidity	DUX4	SMCC	TfR1-targeted mAb	DUX4-siRNA	FSHD	Phase III
AOC-1044	Avidity	Exon-44	SMCC	TfR1-targeted mAb	Exon-44-skipping PMO	DMD	Phase III
DYNE-101	Dyne	DMPK	Val-cit	TfR1-targeted Fab	DMPK-ASO	DM1	Phase I/II
DYNE-251	Dyne	Exon-51	Val-cit	TfR1-targeted Fab	Exon-51-skipping PMO	DMD	Phase I/II
TAC-001	Dyne	TLR9	PEG ₂₄ Phosphate bond	CD22-targeted mAb	CpG(TLR9 agonist)	Cancer	Phase I/II

TfR1 transferrin receptor 1, *CD22* Siglec-2, *DMPK* dystrophin myotonia protein kinase, *siRNA* small interfering RNA, *PMO* phosphorodiamidate morpholino oligomer, *DUX4* double homeobox protein 4, *ASO* antisense oligonucleotide, *CpG* cytosine phosphoric acid guanine, *TLR9* toll-like receptor 9, *DM1* myotonic dystrophy type 1, *DMD* Duchenne muscular dystrophy, *FSHD* facioscapulohumeral muscular dystrophy, *SMCC* succinimidyl 4-N-maleimidomethylsyclohexane-1-carboxylate.

group with 2'-O-methyl or 2'-methoxyethyl groups, as well as incorporating nucleotides such as locked nucleic acid, unlocked nucleic acid, or glycol nucleic acid analogs into the siRNA structure [11–13]. This provides a critical foundation for drugability in the subsequent development of AOC technology and propels its evolution toward more efficient and stable therapeutic applications.

In 2022, AOC 1001 (Avidity Biosciences), targeting Dystrophin Myotonia Protein Kinase (DMPK), has entered clinical development, marking a milestone in the field. Leveraging the widespread expression and efficient internalization of TfR1 on muscle cell surfaces, AOC 1001 achieves targeted delivery to muscle tissue. To reduce immunogenicity and avoid off-target toxicity, the molecule employs an effector-function-deficient antibody design and utilizes a non-cleavable linker to ensure siRNA integrity during delivery. Preclinical studies demonstrate that a single dose achieves sustained (~75%) target gene silencing in non-human primates [14]. Currently, AOC 1001 has advanced to Phase III clinical trials for the treatment of Myotonic Dystrophy Type 1 (DM1). As shown in Table 1, the extensive pipelines developed by major companies targeting various rare muscular dystrophies such as FSHD and DMD further underscore the immense application potential of the AOC platform [15]. Recently, Avidity's AOC drug AOC 1044 received FDA Fast Track designation and Orphan Drug status, further fueling enthusiasm for AOC drug development.

Despite this, there are still a number of significant obstacles in the way of the clinical application of AOC technology, and its advancement is still in its infancy. Due to the relatively small pool of potential target receptors, the current strong dependence on a few number of targets, including TfR1, greatly restricts the spread of these therapies to extrahepatic tissues and other indications [16]. Furthermore, significant optimization bottlenecks exist in the complex molecular design of these agents: the selection of antibodies and the chemical properties of linkers influence the stability, intracellular release efficiency, ultimately efficacy and safety of the conjugated drug. The interactions among these factors require systematic elucidation. Moreover, even with chemical modifications to enhance stability and affinity, the comprehensive regulatory mechanisms governing the in vivo fate and function of intact AOC drugs remain to be thoroughly explored [17, 18].

This paper systematically elaborates on the core design elements of AOC drugs as illustrated in Fig. 1. First, we emphasize the decisive role of target endocytosis efficiency and analyze the impact of linker chemistry on drug metabolism and efficacy. Then we explore the optimization of delivery efficiency through nucleic acid modifications including PEGylation. Rational selection and engineering strategies for different antibody formats such as IgG, Fab, and bispecific antibodies are also been proposed. Finally, this paper outlines the platform's potential applications in delivering diverse oligonucleotide types, aiming to provide a systematic reference for the rational design and development of AOCs.

MOLECULAR CHARACTERISTICS OF AOCs

AOC combines oligonucleotides with gene regulatory functions to targeted antibodies through covalent linkage, aiming to achieve efficient intracellular delivery of oligonucleotides to specific cells for precise regulation of target gene expression. Structurally, AOC resembles antibody-drug conjugates (ADCs), both comprising three components: an antibody, a linker, and an active payload. However, due to fundamental differences in delivery objectives and payload properties, AOC exhibits distinct design principles compared to ADCs. We systematically compare the design differences between the two conjugates as shown in Table 2.

The core function of AOC lies in utilizing antibodies to deliver gene-regulating oligonucleotides into cells, enabling precise regulation of gene expression at the mRNA level. This fundamentally differs from traditional cytotoxic ADCs, which deliver cell toxins to directly kill target cells. The target receptors of AOC prioritize endocytic efficiency and high circulation properties over mere specificity. The antibody-linker combination must balance specific binding affinity with endosomal escape capability—a critical factor determining gene silencing efficacy. Chemical modifications to the oligonucleotide enhance stability and modulate immunogenicity while preserving long-acting therapeutic properties. Therefore, the creation of AOC drugs is a precision-engineered system that integrates gene regulation, endosomal escape, and targeted endocytosis.

TARGETED RECEPTORS AND THERAPEUTIC INDICATIONS

The therapeutic field selection for AOCs is fundamentally determined by the tissue distribution specificity and endocytosis kinetics of their target receptors. A mature target receptor is crucial for the successful development of AOCs, serving not only as a “navigation system” for precise oligonucleotide delivery to target cells but also directly defining the drug's potential indications. The therapeutic potential of antibody-nucleic acid conjugates in a variety of disease areas, from hematologic malignancies and solid tumor immunotherapy to hereditary muscle disorders and liver diseases, has been preliminarily demonstrated by current research and application of receptors like TfR1, ASGPR, CD22, and SIRPα. Interestingly, AOCs' payload activity goes beyond conventional gene silencing to include immune activation, splicing regulation, and other modalities. Nevertheless, the target tissues that the antibody moiety defines continue to determine their effective function. Therefore, the logical design of AOCs depends on an awareness of the inherent link between “receptor-tissue-disease”. This chapter will explore this fundamental link, highlight important receptor targets that are currently being studied, and show how a mature, validated receptor can act as a tailored delivery platform for AOCs, propelling drug development in a variety of therapeutic domains.

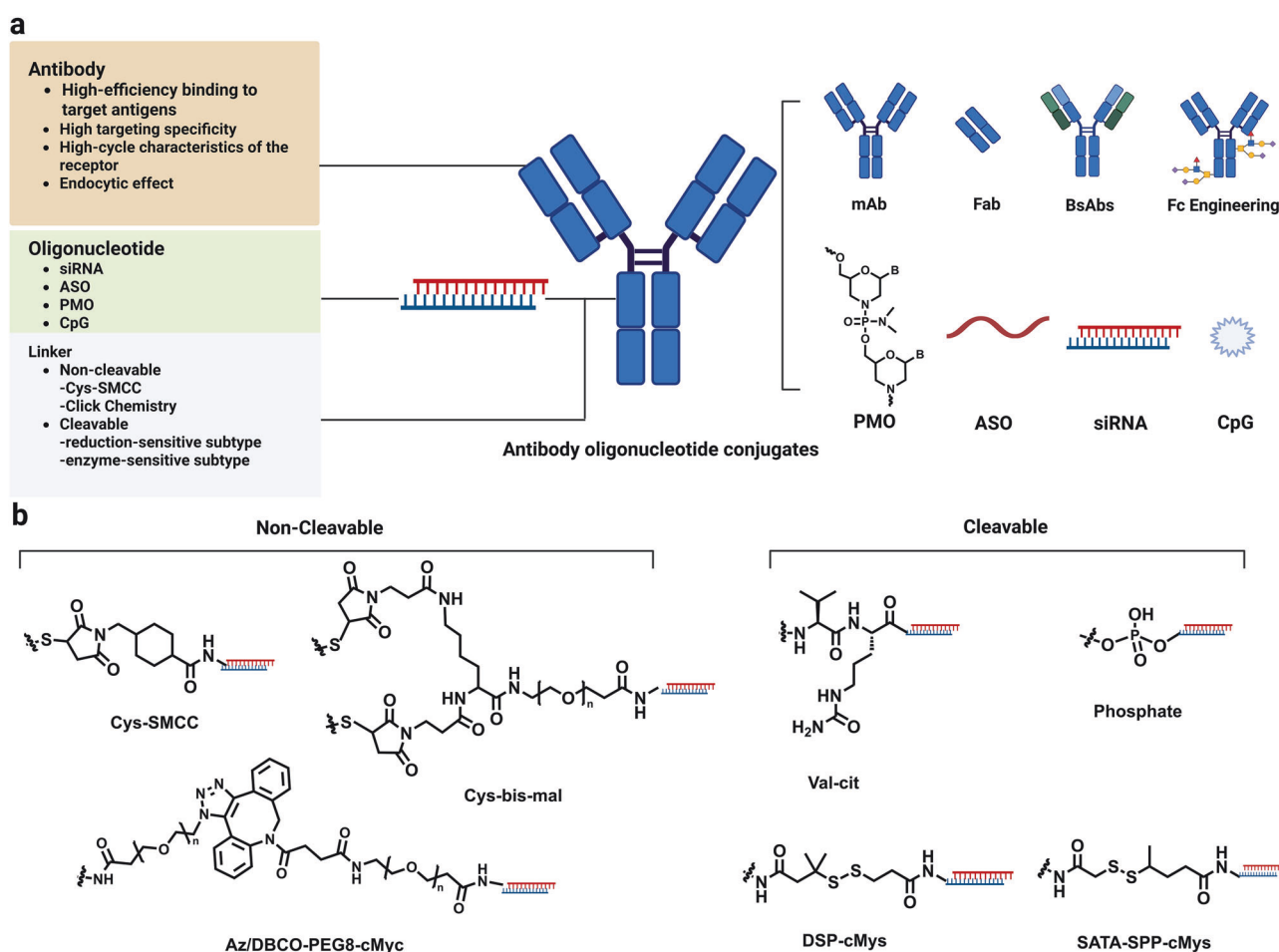


Fig. 1 Molecular architecture of AOCs. a Schematic structure of an AOC. **b** Chemical structures of representative non-cleavable and cleavable linkers used in AOC design. Created in BioRender. Fu, Y. (2026) <https://BioRender.com/owb3b2i>.

TfR1: a multifunctional platform delivery portal

Transferrin receptor 1 (TfR1 or CD71) is a type II transmembrane glycoprotein of approximately 90 kDa. It forms homodimers with a molecular weight of about 180 kDa via disulfide bonds on the cell surface. Each monomer consists of an extracellular domain, a transmembrane domain, and an intracellular tail region [19]. TfR1 is a high-affinity circulating receptor widely expressed on the surfaces of proliferative cells such as tumor cells, myofibers, cardiomyocytes, and cerebral microvascular endothelial cells. Its primary physiological function is to facilitate intracellular iron transport by cooperating with transferrin through clathrin-mediated endocytosis [20]. Following binding to TfR1, antibodies can be co-internalized into endosomes and undergo lysosome-directed transport, ultimately releasing their carried oligonucleotide payloads. Subsequently, TfR1 itself can recycle to the cell membrane surface for the next round of endocytosis. This highly efficient recycling mechanism provides the molecular basis for the repeated delivery of AOCs [19].

In skeletal muscle cells, persistent overexpression of TfR1 offers an effective delivery window for conditions like DM1 and DMD. Avidity Biosciences' lead candidate AOC 1001 successfully corrects splicing defects in the DMPK gene by delivering siRNA targeting DMPK via a monoclonal antibody that targets TfR1. The AOC 1001 mechanism diagram is shown in Fig. 2. Phase I clinical trial data show an increase of over 40% in the coverage of target exons in patients [14]. Instead of using full-length antibodies, Dyne Therapeutics has adopted a new strategy by conjugating Fab fragments that target TfR1. Phase I/II clinical studies have begun for its pipeline medications DYNE-101 (an ASO targeting DMPK for

DM1) and DYNE-251 a PMO targeting DMD exon 51 for DMDs [21, 22].

Due to the high expression of TfR1 in brain microvascular endothelial cells, it is considered an ideal gateway for mediating drug transport across the blood-brain barrier (BBB). Denali Therapeutics developed an AOC targeting TfR1 using its proprietary Antibody Transport Vector (ATV) platform. Preclinical studies demonstrated a 50% reduction in target mRNA expression in mouse cortex and spinal cord, with brain drug concentrations exceeding those of conventional ASOs by more than tenfold [23]. Similarly, Eli Lilly's proprietary data confirm [24].

TfR1, as a clinically validated high-efficiency delivery platform, establishes the foundation for repeated delivery of oligonucleotide therapeutics through its efficient internalization-recycling mechanism. The receptor's specific expression in diverse tissues, including skeletal muscle and the blood-brain barrier, enables it to serve as a single targeting gateway for successful application across multiple therapeutic areas, such as neuromuscular and central nervous system disorders. This fully demonstrates the platform's value of "one receptor, multiple applications".

ENT2: a transporter-based approach for non-endocytic delivery

Unlike TfR1-targeting strategies centered on endocytosis mechanisms, delivery systems targeting the membrane transporter equilibrative nucleoside transporter 2 (ENT2) represent an entirely new delivery paradigm independent of endocytosis [25]. ENT2 is a bidirectional transmembrane transporter widely expressed in

Table 2. Comparison of ADC and AOC characteristics*.

	AOC	ADC
Antibody Design	High-efficiency binding to target antigens. High targeting specificity. Endocytic effect. High-cycle characteristics of the receptor.	High tumor specificity [109]. Rapid internalization. Low immunogenicity. Long plasma half-life [110].
Linker	Stable binding controlled release. Biocompatibility. Facilitates carrier escape from the endoplasmic reticulum into the cytoplasm [111].	Must remain stable in the bloodstream but can be specifically released within tumor cells [112, 113].
Mechanism of Action	siRNA: Cleavage of mRNA, highly efficient and specific gene precipitation [114]. ASO: Antisense hybridization, gene silencing/splicing modification [83, 101]. PMO: Steric hindrance, exon skipping/translation inhibition [115]. CpG: TLR9 activation, immune activation, cancer immunotherapy adjuvant [116, 117].	Topoisomerase I inhibitors: SN38, camptothecin derivatives, etc[118]. Microtubule inhibitor: MMAE [47]. DNA Damage Category: Calicheamicin [119], Pyrrolobenzodiazepines.
Targeted Receptor	Receptors possessing efficient endocytosis capabilities and effective intracellular release mechanisms, such as TfR1, CD22, SIRP α , and ENT2 [120].	Select antibodies targeting highly tumor-specific antigens, such as HER2, Nectin-4, FR α , EGFR, BCMA, and the CD family, to enhance endocytic bystander effects [121].
Primary Application Areas	Genetic disorders, metabolic disorders, neurodegenerative diseases, etc [122].	Oncology [48].
Dosage interval	The persistence of gene regulation and improved pharmacokinetics hold promise for achieving longer dosing intervals (e.g., quarterly or semi-annually).	To maintain sufficient cytotoxic pressure, frequent intravenous administration is required [123].
Safety	Immunogenicity, target-related toxicity (genetic interference with normal tissues expressing the target), liver/kidney accumulation toxicity, immunogenicity.	Off-target toxicity, payload-related toxicity, immunogenicity [124].
Representative drugs	AOC-1001 (Phase III) [19]. AOC-1044 (Obtained FDA Fast Track designation and orphan drug designation) [90].	Enhertu TM , Adcetris TM , Elahere TM , Zynlonta ^{TM36} [125, 126].

*:ADCs represent a broader definition encompassing antibody-drug conjugates carrying biologically active payloads. AOCs essentially constitute a subset of broad-spectrum ADCs, specifically referring to oligonucleotide conjugates. The two are not mutually exclusive. To more clearly illustrate their distinction, the term ADC here refers to traditional cytotoxic antibody-drug conjugates.

multiple tissues (such as skeletal muscle, brain, kidney, and various tumor cells). It primarily mediates the equilibrium transport of purine/pyrimidine nucleosides and nucleobases across cell membranes, while also facilitating the cellular uptake of various nucleoside analog drugs [26, 27]. Research indicates that over-expression of ENT2 is closely associated with disease progression in multiple malignant tumors, making it an attractive therapeutic target [28].

Based on the unique biological function of ENT2, Gennao Bio has developed an innovative GMAB (Genetically Modified Antibody) platform technology. The core of this technology lies in utilizing the genetically engineered anti-DNA antibody 3E10 [29]. 3E10 can enter cells via a non-endocytic pathway with the aid of the nucleoside transporter ENT2, which is highly expressed in tumor cells [30]. Its mechanism of action disrupts the traditional receptor-mediated endocytosis (RME) pathway relied upon by conventional AOCs. By entering cells via the ENT2-mediated non-endocytic pathway, 3E10 efficiently bypasses the endocytosis-endosome-lysosome pathway, fundamentally circumventing the major bottleneck of endosomal escape. Furthermore, 3E10 itself exhibits tumor cytotoxicity. When used to deliver nucleic acid drugs with tumor-killing activity, its antitumor efficacy may be further enhanced.

Humanized 3E10 cell penetration does not depend on the endocytic uptake pathway but is highly dependent on the function of the ENT2 nucleoside transporter. ENT2 is present on both the plasma cell membrane and the nuclear membrane [27], and thus likely facilitates the transmembrane transport of 3E10. ENT2 is also highly overexpressed in multiple human cancers, driven by the high proliferative and metabolic demands of tumor

cells. This unique uptake pathway is thus one of the determining factors [31, 32].

As the platform's first clinical candidate, GMAB-7001 is an innovative linker-free AOC drug whose mechanism of action is illustrated in Fig. 3. Its design further simplifies the structure, potentially enabling direct integration of oligonucleotides into the antibody sequence via genetic encoding to maximize delivery efficiency and product homogeneity. Preclinical data demonstrate its exceptional therapeutic potential: in a pancreatic cancer mouse model, GMAB-7001 significantly enhances lymphocyte infiltration within the tumor microenvironment including multiple key immune cell subsets such as CD45⁺, CD8⁺, CD4⁺, and CD19⁺ cells as illustrated in Fig. 3, indicating its capacity to activate anti-tumor immune responses. The drug also exhibits superior central nervous system (CNS) penetration, effectively reducing intracranial tumor burden by 50% and inhibiting spinal metastasis formation, demonstrating significant potential for treating brain tumors and metastatic lesions [30].

ENT2-targeted AOC represents an innovative strategy distinct from mainstream approaches. By leveraging the function of membrane transporters rather than the properties of endocytic receptors, it offers a novel solution for oligonucleotide delivery. This approach demonstrates unique advantages, particularly in overcoming the challenge of escaping the endosomal compartment and targeting the nervous system—a tissue notoriously difficult to internalize. However, its future development requires close attention to potential off-target toxicity risks associated with ENT2's widespread expression, and the detailed molecular pathways underlying this novel “direct transport” mechanism remain to be further elucidated.

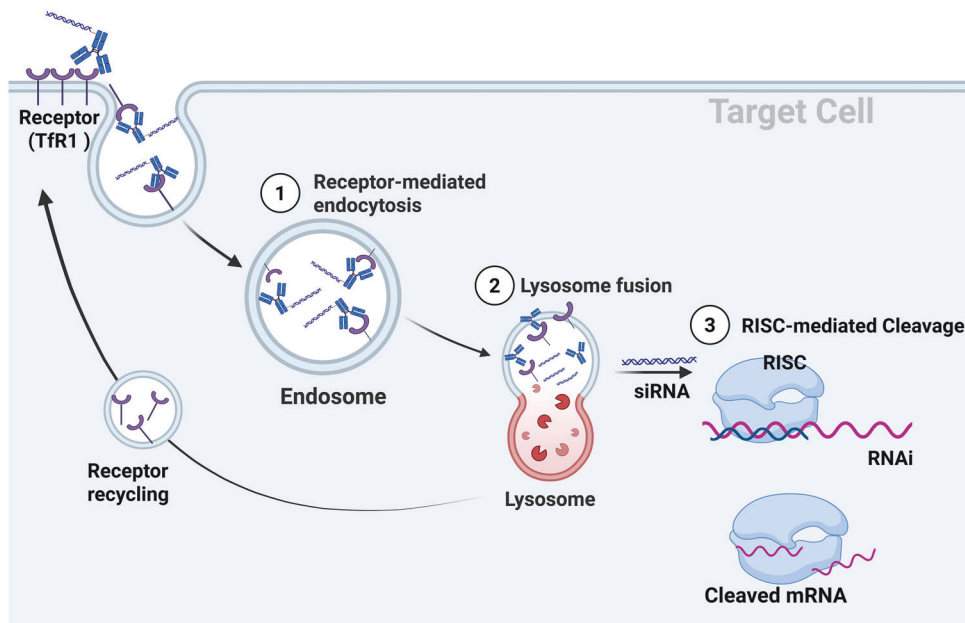


Fig. 2 Schematic diagram of the mechanism of action of AOC-1001. Adapted from [8].

Targeting immunomodulatory receptors: precision reprogramming of CD22 and SIRPα signaling

In tumor immunotherapy, a core strategy for precise immune reprogramming involves targeting inhibitory receptors on immune cells to block immunosuppressive signals while delivering specific activation stimuli [33]. CD22 and signal-regulated protein alpha (SIRPα) exemplify this approach: CD22 is expressed specifically on B cells, and SIRPα is restricted to myeloid cells. Their confined expression patterns and efficient endocytic capacity make them ideal targets for AOCs.

CD22 is a critical inhibitory receptor on the surface of B cells. It negatively regulates B cell receptor (BCR) signaling via the immunoreceptor tyrosine-based inhibitory motif (ITIM) within its intracellular domain, restraining aberrant B cell activation. Its sustained overexpression in more than 90% of B-cell malignancies, coupled with robust endocytic activity, renders it a superior target for intracellular payload delivery [34–37]. SIRPα, also known as SHPS-1/BIT/CD172a, is an immunoglobulin superfamily protein that binds to protein tyrosine phosphatases SHP-1 and SHP-2 via its cytoplasmic domain. Another immunoglobulin superfamily protein, CD47, serves as the ligand for SIRPα, and together they form an intercellular communication system (the CD47-SIRPα signaling system). SIRPα is primarily expressed on myeloid cells (e.g., macrophages, dendritic cells). Upon binding its widespread ligand CD47, it transmits a potent “don’t eat me” signal that inhibits phagocytosis, serving as a key mechanism for tumor cells to achieve immune evasion. Blocking this pathway can reverse the suppression of macrophage function, but its broad expression profile also poses challenges such as hematotoxicity [38–42].

The innovative TRAAC platform, developed by Tallac Therapeutics, capitalizes on the unique biology of two distinct immune targets through a modular antibody-conjugate strategy. The core technology of TRAAC involves site-specific conjugation of potent Toll-like receptor 9 (TLR9) agonists—CpG oligodeoxynucleotides—to targeting antibodies via optimized cleavable linkers, enabling precise delivery of immunostimulatory payloads to designated immune cell populations.

The lead candidate, TAC-001, demonstrates this approach by targeting CD22 to facilitate efficient CpG internalization into B cells. Subsequent intracellular TLR9 activation drives dual mechanisms: direct stimulation of antitumor B-cell functions and initiation

of systemic antitumor immunity through enhanced antigen presentation and cytokine production. Preclinical studies indicate potential of TAC-001 to overcome resistance to conventional immune checkpoint inhibitors. The companion candidate, ALTA-002, applies the same platform principle to target SIRPα on tumor-associated myeloid cells. This bispecific molecule achieves coordinated immune activation through two complementary pathways: robust TLR9 stimulation via targeted CpG delivery concurrent with blockade of the inhibitory SIRPα-CD47 axis. This dual action reprograms immunosuppressive myeloid populations—particularly M2-polarized macrophages—toward pro-inflammatory, phagocytically active states, ultimately priming robust T-cell-mediated antitumor immunity through enhanced antigen cross-presentation and epitope spreading, as schematically summarized in Fig. 4.

By merging targeted delivery with immune activation, AOC technology establishes a new paradigm that extends its functionality beyond gene silencing to encompass immune modulation. Capitalizing on its precise cellular selectivity, this approach strives to achieve an optimal therapeutic index by navigating the critical balance between efficacy and immune-related safety. This positions AOCs as a promising strategy in cancer immunotherapy, especially for tackling refractory malignancies and overcoming immunosuppressive tumor microenvironments.

ANTIBODY ENGINEERING DESIGN IN AOC

The efficacy of AOCs largely depends on the rational design of their antibody moieties. In AOC design, the antibody moiety is far from a simple targeting tool; the choice of its form, such as full-length IgG or Fab fragments, constitutes a core strategic decision that profoundly impacts the drug’s targeting efficiency, pharmacokinetic behavior, safety profile, and ultimate therapeutic efficacy. The choice of antibody format requires careful balancing based on four key factors: payload characteristics, the biological microenvironment of the target tissue, the desired pharmacological effect, and the risk of immunogenicity [43]. Currently, mainstream strategies focus on two primary technical approaches: one involves utilizing full-length IgG antibodies that have undergone engineering modifications, while the other employs smaller antigen-binding fragments. Taking Avidity Biosciences and Dyne

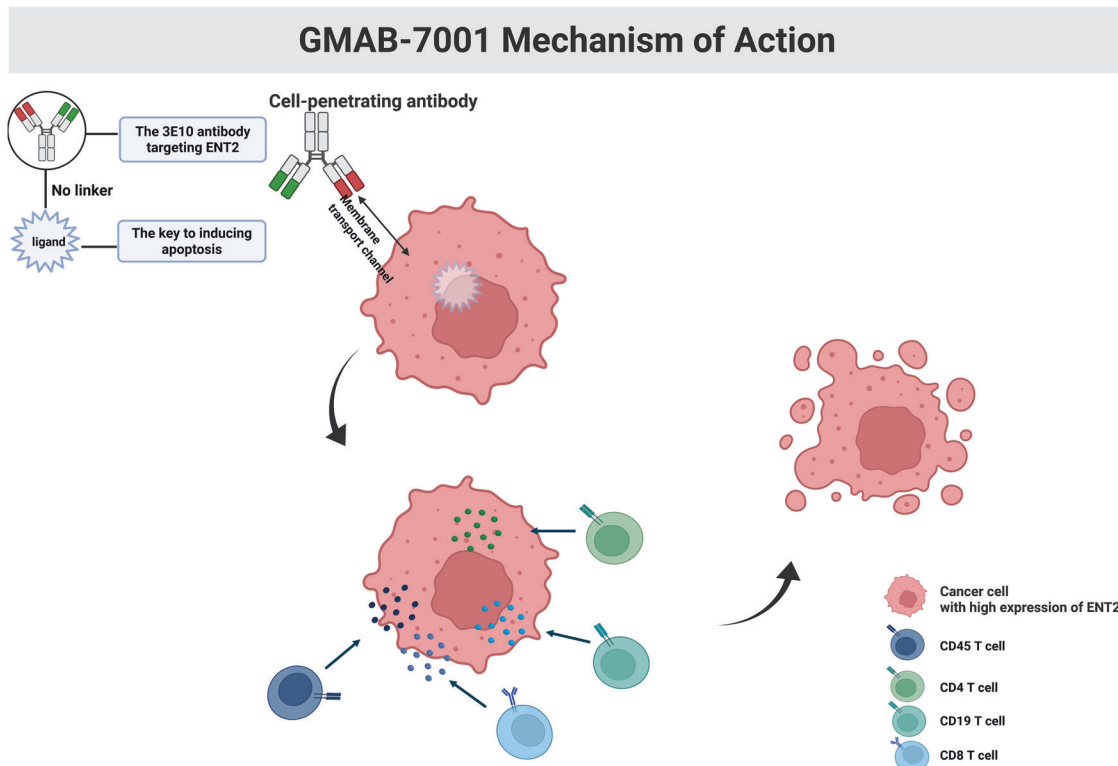


Fig. 3 Schematic diagram of the mechanism of action of GMAB-7001. GMAB-7001 targets the ENT2 receptor and utilizes ENT2's intrinsic membrane transport channel to significantly enhance lymphocyte infiltration in the tumor microenvironment (including multiple key immune cell subsets such as CD45⁺, CD8⁺, CD4⁺, and CD19⁺ cells), demonstrating its capacity to activate antitumor immune responses. Adapted from [25, 127].

Therapeutics—both targeting Tfr1—as examples, the starkly different antibody design choices made by these two companies provide an excellent comparative case study for understanding the rational design of AOCs.

Full-length IgG strategy

Avidity's AOC-1001 utilizes a fully humanized, full-length IgG1 antibody. The core design rationale of this strategy is that the Fc region of the full-length IgG effectively binds to FcRn, mediating the classic pH-dependent recycling process. This avoids lysosomal degradation, enabling the antibody to maintain a prolonged half-life in the systemic circulation (typically lasting several weeks). This enables extremely high systemic exposure following a single dose, delivering sustained therapeutic effects. It provides the foundation for reducing dosing frequency and improving compliance in chronic disease patients [44, 45]. Its larger molecular weight also limits non-specific tissue penetration. Regarding targeting, this strategy relies on the antibody's high-affinity binding to Tfr1 to achieve targeting of the target cells. As for immunogenicity, despite humanization, the large protein mass and complex structure of the full-length antibody still potentially carry a relatively high immunogenicity risk. Clinical data for AOC-1001 reported the development of anti-drug antibodies (ADA) in some patients, with the most common adverse reactions being nausea and headache. These may be associated with immune responses or broad target binding.

Fab fragment strategy

Dyne Corporation's DYNE-101 and DYNE-251 employ a Fab fragment strategy targeting Tfr1. Compared to full-length antibodies, Fab fragments weaken the interaction strength with Tfr1, preventing significant receptor downregulation and degradation. This preserves the receptor's circulating endocytosis

capacity, offering the potential for more efficient and reproducible oligonucleotide delivery [46]. Unlike IgG strategies, Fab fragments possess smaller molecular weights and lack the Fc domain, preventing them from extending their half-life through FcRn-mediated recycling pathways. Consequently, their serum half-life is significantly reduced (typically ranging from hours to days), resulting in lower systemic exposure [47]. This characteristic does indeed necessitate more frequent dosing, but its advantage lies in rapid clearance, which helps minimize systemic exposure and associated toxicity risks—making it particularly suitable for treatment scenarios sensitive to systemic toxicity. Additionally, the smaller size may enhance tissue penetration, offering value especially in therapeutic applications requiring deep penetration into muscle tissue [48, 49]. Structurally, Fab fragments are simpler and theoretically pose a lower immunogenicity risk than full-length antibodies. Preliminary clinical results indicate that DYNE-101 exhibits a lower incidence and severity of treatment-related adverse events, reflecting the potential safety advantages of this strategy.

Diversified antibody engineering strategies

Beyond the delivery strategies based on the two classical antibody formats, AOC development is now exploring more complex antibody engineering solutions to expand its therapeutic potential. Among these, bispecific antibodies enhance tissue selectivity through cooperative targeting, while Fc-engineered variants allow for fine-tuned pharmacokinetics and immune effector functions, together providing key insights for the design of AOCs.

BSABS

Bispecific antibodies (BsAbs) refer to genetically engineered artificial antibodies capable of simultaneously binding two distinct

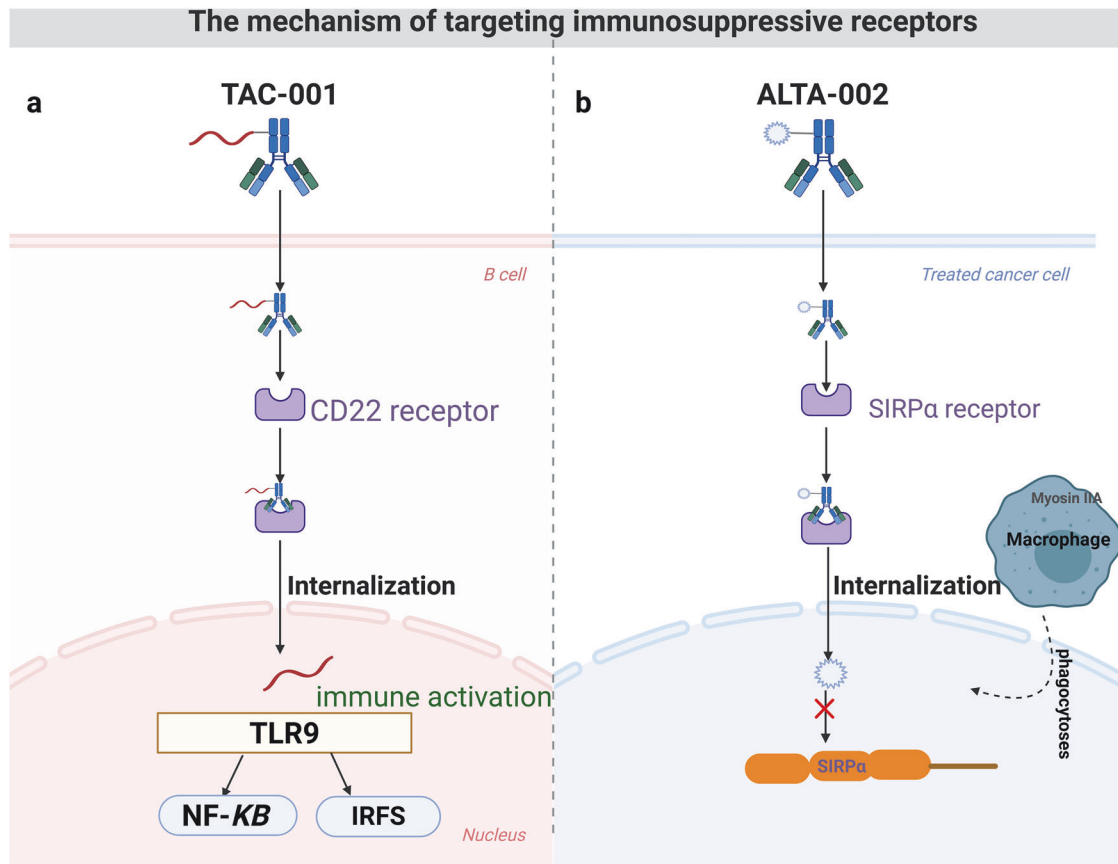


Fig. 4 Schematic diagram of the mechanism targeting immunosuppressive receptors. a TAC-001, an ISAC targeting CD22, carries a chemically modified CpG TLR9 agonist. **b** ALTA-002, an ISAC targeting the SIRP α receptor, delivers its loaded CpG to myeloid cells in the tumor microenvironment, thereby locally blocking the SIRP α -CD47 inhibitory signal. Adapted from [128].

antigens or different epitopes of the same antigen. These antibodies overcome the limitation of traditional monoclonal antibodies, which are specific to a single target, offering new directions for AOC design through their precise dual-targeting capability [31]. A common application strategy for BsAbs in AOCs involves constructing “conditionally activated” systems: for instance, one arm targets a widely expressed endocytic receptor (such as TfR1) to mediate tissue penetration and cellular uptake; while the other arm highly specifically binds to a lesion-specific marker (e.g., β -secretase BACE1), ensuring oligonucleotide release primarily within target cells. This significantly enhances delivery precision and reduces off-target toxicity [32].

In terms of central nervous system delivery, AOCs represented by anti-BACE1/TfR1 bispecific have demonstrated significant potential. In preclinical studies, the bispecific antibody developed by Benjamin A. Smith et al. demonstrated that TfR1-mediated transport across the blood-brain barrier substantially enhances the distribution of the BACE1 antibody within the brain. This bispecific antibody significantly increases brain tissue uptake and sustainably reduces intracerebral A β levels following a single administration, achieving long-lasting inhibition of Alzheimer’s disease-related targets [50]. BsAbs can effectively promote the targeted delivery of oligonucleotides to hard-to-reach tissues such as the brain. Furthermore, BsAbs can be employed in combination therapies, such as simultaneously binding tumor antigens and immune cell receptors, thereby activating immune responses while delivering gene-silencing drugs to achieve a dual therapeutic effect [51–53]. Despite the significant advantages of BsAbs-AOCs in precision delivery and synergistic therapy, BsAbs-AOCs still face challenges such as complex molecular structures,

difficult conjugation processes, immunogenicity risks, and unconventional pharmacokinetic behavior. Currently, this field remains in the research and development stage, with some candidate drugs already entering clinical trials. Future efforts should focus on further optimizing antibody design, conjugation strategies, and production processes to advance their translation into clinical applications.

FC ENGINEERING

Antibody variants obtained through genetic engineering of the Fc fragment are fundamentally designed to precisely modulate binding characteristics to Fc γ receptors (Fc γ Rs) or neonatal Fc receptors (FcRn) via amino acid substitutions or glycosylation modifications [54]. Mutating the Fc region to enhance its affinity for FcRn can further prolong the half-life of IgG. For example, variants developed by Sang et al. can extend the half-life by 2–3 times [55]. Conversely, modifications can also reduce FcRn binding to achieve rapid clearance. The goal of Fc engineering is precisely to achieve “fine-tuned regulation” of antibody half-life, signifying that half-life is no longer a fixed attribute but a variable that can be actively designed according to clinical needs. Furthermore, targeted modifications to the Fc region can enhance the antibody’s specific recognition and endocytosis efficiency toward target cells, thereby improving the effective delivery of oligonucleotides [56–58]. Consequently, engineered Fc variants provide a multifunctional regulatory module for rational AOC design, enabling synergistic optimization across pharmacokinetics, safety, and targeting efficiency. Ionis Pharmaceuticals is exploring smaller TfR1-binding agents such as camelid single-domain antibodies

and bicyclic peptides [59], pursuing extreme miniaturization and efficient delivery.

Antibody engineering is the strategic core of AOC drug design, where form selection inherently involves a precise balancing act across three dimensions: targeting, pharmacokinetics, and safety. Full-length IgG, with its extended half-life, provides the foundation for managing chronic diseases requiring sustained systemic exposure. Conversely, Fab fragments leverage superior tissue penetration and rapid clearance for localized, efficient delivery and toxicity control. Neither approach is inherently superior; their core differences reflect distinct design philosophies tailored to varying clinical needs. Concurrently, bispecific antibodies significantly enhance targeting precision through “logic gating” while Fc engineering enables “programmable” fine-tuning of critical properties like half-life. Moving forward, AOC antibody design will transcend the limitations of single-form choices and evolve toward intelligent, scenario-specific customization based on AI-driven prediction and modular assembly. This approach will deliver truly precise and tunable delivery solutions for diverse extrahepatic diseases.

LINKERS

The linker serves as the molecular bridge for AOC, with its chemical properties, stability, and release mechanism directly determining the pharmacokinetic behavior, biodistribution, therapeutic efficacy, and safety of AOC [60]. Based on their coupling properties, AOC linkers are primarily categorized into two major types: cleavable linkers and non-cleavable linkers. Cleavable linkers are designed to exploit differences between the intracellular microenvironment of target cells and the external circulation for specific activation. They primarily include pH-sensitive linkers such as hydrazone, reduction-sensitive linkers such as disulfide bonds [17, 61], and enzyme-sensitive linkers, particularly dipeptide-based linkers. Although the canonical dipeptide sequences (e.g., Val-Cit) were specifically designed and optimized for efficient cleavage by cathepsin B, they are also susceptible to processing by other lysosomal proteases in vivo [62]. The greatest risk associated with cleavable linkers is their potential to undergo specific hydrolysis, reduction, or enzymatic degradation in the bloodstream, leading to off-target effects, shortened half-lives, and other adverse consequences. Non-cleavable linkers such as thioether bonds and maleimide hexanoyl bonds do not undergo cleavage within cells. Their release depends on the complete proteolytic degradation of the antibody carrier within lysosomes, thereby releasing oligonucleotides still connected to amino acid residues. These linkers minimize off-target release but their efficacy critically depends on the released oligonucleotide-amino acid complex retaining functionality and possessing effective endosomal escape capabilities [62]. Maleimide linkers are frequently employed in such designs due to their excellent stability and moderate conformational flexibility.

Enzyme-cleavable linkers, particularly cathepsin B-sensitive dipeptides such as Val-Cit, are widely used in AOCs to enable controlled intracellular oligonucleotide release. This strategy has shown significant promise in preclinical models of DM1. For example, the AOC candidate DYNE-101 (DTX-C-008), which features a Val-Cit linker site-specifically conjugated to an engineered anti-TfR1 antibody, mediated efficient and muscle-specific delivery in vivo. Administration at 3 mg/kg led to sustained DMPK mRNA reduction, which reached 50% in gastrointestinal muscle and 47% in tibialis anterior, with effects persisting for 28 days, and no detectable activity in off-target tissues such as spleen or brain. In contrast, the related conjugate DTX-C-007, which uses a Val-Arg linker with an IgG2a Fab, showed no efficacy, underscoring that AOC potency depends critically on synergistic compatibility between the linker and antibody architecture [63]. Similarly, the DMPK-targeting AOC-1001 also

employs a non-cleavable SMCC linker. At a dose of 3 mg/kg, it achieved up to 80% DMPK knockdown in skeletal muscle while exhibiting similarly sustained activity, confirming that diverse linker chemistries can support effective muscle-directed delivery [14]. In an exon-skipping application, DYNE-251 delivered a PMO using an anti-TfR1 Fab conjugated via a Val-Cit linker. Administration at 30 mg/kg restored dystrophin expression to over 60% of wild-type levels and reaching 90% in the diaphragm in a DMD mouse model four weeks post-treatment, further underscoring the platform's capacity to restore functional protein expression [64, 65].

The functional performance of linkers is governed by their physicochemical properties. For cleavable linkers, key stabilization strategies include chemical modifications, such as gem-dimethyl substitution in disulfide bonds, and sequence optimization to fine-tune cleavage kinetics. In general, shorter linkers tend to exhibit superior metabolic stability. For instance, AOCs incorporating shorter disulfide linkers have demonstrated significantly improved stability in mouse whole blood compared to extended analogs, thereby better preserving the average (drug-to-antibody ratio) [66]. Linker length and conjugation site also substantially influence AOC bioactivity, partly by modulating the overall steric conformation of the conjugate [67]. These parameters can affect antibody-antigen engagement, oligonucleotide interaction with effector complexes such as RISC, as well as payload release kinetics and oligonucleotide integrity. Experimental evidence indicates that conjugating a non-cleavable linker to the 3' or 5' end of the siRNA antisense strand has minimal impact on RISC loading, whereas internal modifications to the sense strand can markedly compromise in vivo activity [68]. These observations underscore that in AOC design, linker placement is as critical as linker chemistry, necessitating systematic empirical optimization.

OLIGONUCLEOTIDE ENGINEERING: EVOLVING FROM STABILIZATION TO FUNCTIONALITY

In AOC design, chemical modifications of the oligonucleotide are crucial for its stability, specificity, and activity, and must be coordinated with linker chemistry to regulate intracellular release kinetics, thereby ensuring the overall structural stability of the antibody-oligonucleotide conjugate. Furthermore, the role of PEG in AOCs has evolved from a basic conjugation function to a key engineered component capable of actively modulating the in vivo distribution and pharmacokinetics of the oligonucleotide, highlighting the highly integrated and rationalized nature of AOC molecular design.

Chemical modifications: a strategy for targeted delivery

Due to their distinct delivery mechanisms, AOCs and GalNAc-oligonucleotide conjugates employ fundamentally divergent chemical modification strategies as illustrated in Fig. 5. GalNAc-oligonucleotide conjugates enable hepatocyte-specific delivery via multivalent N-acetylgalactosamine ligands that target the asialoglycoprotein receptor [69, 70]. Chemically modifying the oligonucleotide component, which contains phosphorothioate backbone stabilization, 2'-O-methyl or 2'-fluoro sugar substitutions, and other alterations, serves to improve nuclease resistance, extend plasma half-life, promote tissue uptake via protein binding, and fine-tune pharmacokinetic properties. Incorporating hydrophobic groups such as cholesterol can further influence intracellular trafficking and may support endosomal escape [71, 72]. GalNAc-siRNA conjugates commonly use 2'-OMe and 2'-F modifications to balance stability with efficient RISC loading, whereas GalNAc-ASOs frequently employ sterically hindered groups such as 2'-methoxyethyl or constrained ethyl to enhance hybridization affinity. While PEGylated antisense oligonucleotides typically show tissue half-lives of 2–3 days, versions incorporating sugar modifications such as MOE achieve half-lives of several weeks, underscoring how

chemical design governs oligonucleotide persistence in vivo [73, 74].

In contrast, the chemical modification strategy for antibody-oligonucleotide conjugates must adapt to a more complex targeted delivery paradigm. The core design challenge lies in preserving the structural integrity of the conjugate while synergizing with linker chemistry to finely tune intracellular release kinetics. This dual requirement, ensuring plasma stability and enabling controlled intracellular activation, sets AOC optimization apart from the simpler hepatic-targeting strategy of GalNAc conjugates [75, 76].

AOCs employ a distinct design logic for their oligonucleotide components, even when using conventional stabilizing modifications such as phosphorothioate (PS) backbones and 2'-OMe/2'-F sugar substitutions. First, all chemical modifications and conjugation sites must avoid the antibody complementarity-determining regions (CDRs) to maintain antigen-binding activity. Second, the inherent targeting specificity of the antibody eliminates the need for additional targeting moieties such as cholesterol, which could otherwise disrupt pharmacokinetics or promote off-target interactions. Moreover, nucleotide modifications should be coordinated with cleavable linkers—such as pH-sensitive or enzyme-cleavable motifs—to promote efficient endolysosomal release and functional activity of the oligonucleotide payload. This modular, antibody-guided delivery approach underpins the versatility of AOCs and their therapeutic potential in targeting extrahepatic tissues, including cardiac, renal, muscular, and central nervous systems.

PEG modification: from pharmacokinetic tuning to biodistribution design

Polyethylene glycol (PEG) modification is a well-established strategy for enhancing the pharmacokinetic profile of biotherapeutics. By increasing hydrodynamic volume and providing steric shielding, PEGylation reduces renal clearance, prolongs plasma half-life, protects against enzymatic degradation, and can attenuate the inherent immunogenicity of the therapeutic cargo, although PEG itself has the potential to elicit anti-drug antibodies in some cases [77, 78]. The clinical success of PEGylated therapeutics—such as GLP-1 receptor agonists and interferon alpha—underscores the broad utility of this approach [72, 79, 80].

In oligonucleotide therapeutics, PEG modification exhibits clear structure–activity relationships. PEG molecular weight is positively correlated with efficacy; for example, conjugating a 40 kDa PEG to ASOs provides strong steric shielding, leading to optimal serum stability and extended in vivo activity. The site of PEG attachment also significantly affects biodistribution: internally PEGylated ASOs—with PEG conjugated at a central site—show enhanced

accumulation in tumor tissue compared to terminally PEGylated analogs. In the case of siRNA, an external PEG coating—as used in lipid nanoparticles (LNPs)—effectively protects against nuclease degradation and promotes favorable tissue distribution [68, 81].

These design principles have been successfully incorporated into advanced AOC development. As illustrated in Fig. 6, companies such as Avidity Biosciences (with AOC 1001) and Dyne Therapeutics have integrated PEG segments of optimized molecular weight as linkers or spacers in their conjugate platforms. PEG inclusion significantly prolongs the systemic circulation of AOCs, thereby addressing the central challenge of balancing efficient intracellular release with adequate systemic exposure.

Consequently, the role of PEG in AOCs has evolved substantially—from a conventional stabilizer, as traditionally used in ADCs, to an engineered component that actively modulates the pharmacokinetics and biodistribution of the oligonucleotide payload. This functional elevation underscores the integrated and rational design underlying modern AOC platforms.

VERSATILE THERAPEUTIC APPLICATIONS: DIVERSIFIED OLIGONUCLEOTIDE DELIVERY VIA AOCs

The AOC platform is distinguished by its modular design: a validated tissue-targeting antibody can serve as a universal delivery chassis for oligonucleotide payloads with diverse mechanisms of action. This versatility significantly broadens the therapeutic scope of AOCs [82]. Here, we summarize recent advances in AOC-based delivery of ASOs, PMOs, and immunostimulatory oligonucleotides such as CpG oligodeoxynucleotides (CpG ODNs).

ASO

ASOs are single-stranded DNA molecules 15–20 nucleotides in length that precisely regulate gene expression primarily through RNase H1-dependent degradation or steric hindrance effects (e.g., modulating pre-mRNA splicing) [83, 84]. However, its polyanionic nature and large molecular weight result in poor cellular uptake efficiency, limiting its application in extrahepatic tissues. AOC technology offers a robust solution for the targeted delivery of ASOs. For example, Zhang et al. [59] utilized mirrored nucleic acids (L-DNA) to construct a modular antibody-oligonucleotide conjugate (MAOC) screening platform. This study demonstrated that MAOCs efficiently delivered ASOs into the nucleus and successfully mediated exon skipping in the dystrophin transcript. More significantly, the platform enabled the co-delivery of two distinct ASOs to achieve dual exon skipping in mice, highlighting the

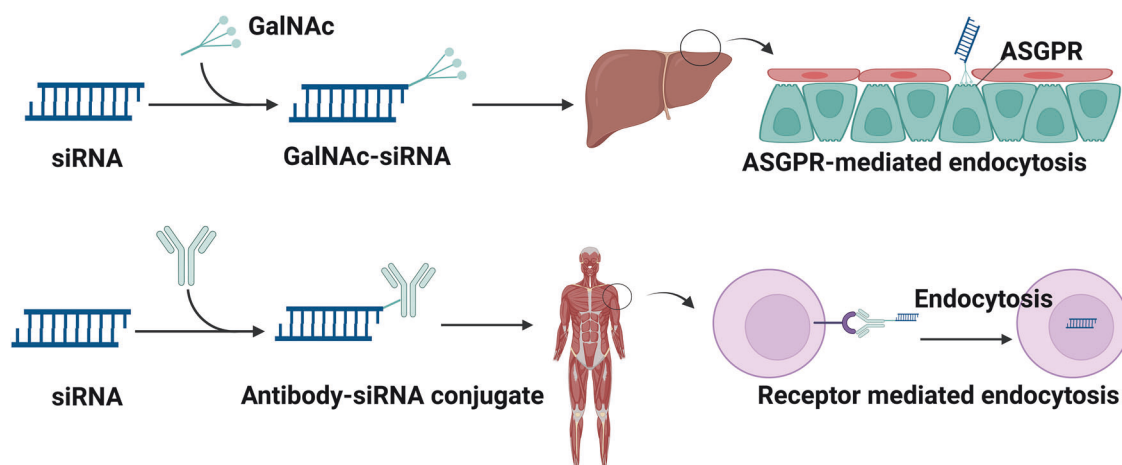
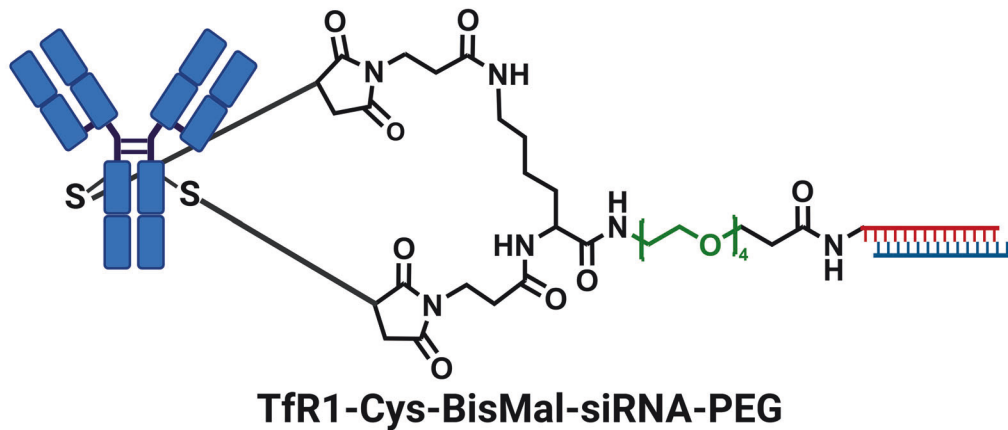
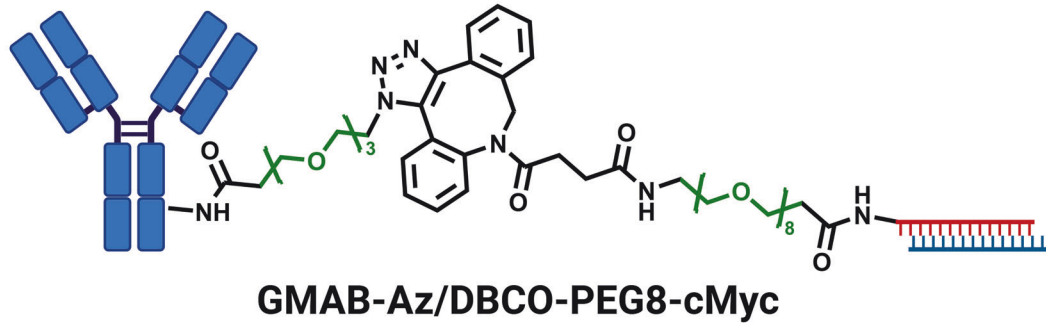


Fig. 5 Mechanism of action of GalNAc-oligonucleotide conjugates and AOCs. Adapted from [129].

a. PEG space linker



b. PEG tail

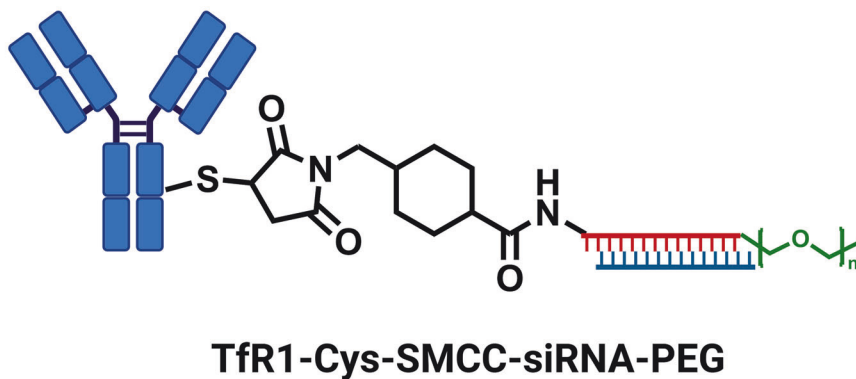


Fig. 6 PEG conjugates currently under clinical investigation. **a** PEG serves as a spacer linker to increase steric hindrance [130, 131]. **b** Terminal PEG conjugation influences pharmacokinetics [130].

immense potential of MAOC platforms for synergistic gene therapy.

PMO

PMO is an electrically neutral antisense oligonucleotide analog with exceptional nuclease resistance. Its backbone consists

of a morpholine ring and phosphoramidite linkages, binding to target mRNA via standard Watson-Crick base pairing. It primarily inhibits translation or modulates splicing through steric hindrance mechanisms [85]. Several PMO-based drugs, such as Eteplirsen for DMD, have already been approved for market release [86–89], but their extrahepatic delivery also faces challenges.

The AOC platform has emerged as a key strategy for overcoming PMO delivery bottlenecks. Avidity Biosciences' AOC 1044 (targeting TfR1 to deliver PMO against DMD exon 44) and Dyne Therapeutics' DYNE-251 (targeting exon 51 PMO) both demonstrated highly efficient and sustained exon skipping effects in preclinical models [90, 91]. These successes not only validate the feasibility of AOC-mediated PMO delivery but also underscore its broad potential for treating inherited muscle disorders.

CPG ODN

The application of AOC has extended beyond gene silencing and regulation into the field of immune activation [92]. CpG ODNs are synthetic DNA molecules containing unmethylated CpG motifs that act as agonists for Toll-like receptor 9 (TLR9). They activate B cells and plasma cell-like dendritic cells, thereby eliciting potent innate and adaptive immune responses. When conjugated with antibodies targeting tumor-associated antigens (such as CD22), they enable precise activation of antigen-specific immune cells, offering a novel strategy for cancer immunotherapy [93].

However, the inherent strong hydrophilicity and polyionic charge characteristics of such single-stranded oligonucleotides readily lead to non-specific interactions with cell membranes, a phenomenon first observed by Walker et al. as early as 1995 [94]. Recent in-depth studies have further revealed that conjugating single-stranded oligonucleotides (e.g., CpG ODNs, ASOs) with antibodies may exacerbate this off-target binding [95]. This finding provides crucial guidance for designing AOCs based on immunostimulatory oligonucleotides, suggesting that conjugation strategies must be meticulously optimized to minimize off-target effects.

CHALLENGES AND FUTURE PROSPECTS

AOC has effectively created a modular "targeting-delivery-regulation" platform by combining the powerful gene regulatory properties of oligonucleotides with the accurate targeting capabilities of antibodies. This discovery opens a new avenue for circumventing oligonucleotide treatments' liver-confined limits and obtaining tissue-specific treatment outside of the liver. As of right now, tactics that target receptors like TfR1 have shown initial viability in clinical settings, which is a crucial step in making this idea a reality.

However, the comprehensive development of AOC still faces a series of core challenges. First, internalization receptor resources are scarce and limited. Current successful cases heavily rely on a very small number of targets such as TfR1 and ENT2, severely restricting their expansion to broader tissues. Even for the established target TfR1, issues like insufficient tissue specificity and off-target uptake persist. This receptor is widely expressed across multiple tissues, making precise targeting difficult. Simultaneously, it is unexpectedly consumed in large quantities by reticulocytes, leading to loss of effective dosage and potential safety concerns. Second, breakthroughs are still needed in the overall delivery efficiency. This encompasses both the classic bottleneck of inefficient escape from endosomes after cellular uptake and the clinical translation uncertainties inherent in the complex chain of events from target binding to the ultimate generation of therapeutic effects. For instance, in indications like DM1, the quantitative relationship between oligonucleotide-mediated splicing correction rates and clinical functional improvement in patients remains incompletely elucidated, posing challenges for efficacy prediction and dose optimization. All of these obstacles limit the ultimate effectiveness and wide-ranging use of AOC technology. In the future, overcoming these obstacles will need deep integration with cutting-edge disciplines and diversified technical pathways. There are three main strategies emerging.

First, antibody engineering is evolving toward intelligence and multifunctionality to enhance targeting precision and address off-target uptake challenges. On one hand, deep engineering of known targets draws inspiration from advancements in the ADC field, where bispecific antibody strategies introduce a new dimension of "logic gating" to AOC design. By simultaneously targeting widely expressed endocytic receptors and disease-specific markers, delivery systems can be engineered to activate exclusively within target cells. This approach enhances selectivity while reducing off-target toxicity. For instance, in central nervous system delivery, AOCs like the anti-BACE1/TfR1 bispecific have demonstrated significant potential in preclinical studies to breach the blood-brain barrier and achieve brain-specific delivery. Meanwhile, novel target discovery platforms and validation examples are expanding delivery frontiers. Manifold Bio's "dual barcode" high-throughput in vivo screening platform, for instance, can simultaneously screen millions of protein variants for tissue distribution. Leveraging machine learning, it directly identifies novel binding molecules with superior tissue targeting based on delivery efficiency data in vivo, aiming to systematically discover novel targeting tools for hard-to-reach tissues. Concurrently, validation of emerging targets has achieved critical milestones. Judo Bio's kidney-targeting delivery platform for glycoprotein receptors has achieved highly efficient, kidney-selective gene silencing in non-human primate models, offering a viable new pathway to overcome limitations of traditional receptors like TfR1. This discovery paradigm, combining data-driven approaches with functional validation, holds promise for continuously delivering next-generation, highly specific targeting receptors to the AOC field.

Second, delivery vehicles keep growing in order to get around endosomal escape and tissue penetration efficiency limits. Beyond antibodies, peptide oligonucleotide conjugates (POCs)—a vital parallel technology—show distinct benefits in overcoming the main obstacle of endosomal escape because of their reduced size and engineering potential [96–98]. Entrada Therapeutics' proprietary cell-penetrating peptide platform comprises circular peptides intended to resist plasma protease breakdown. These peptides attain an endosomal escape rate of over 50% following non-specific binding to cell membranes and endocytosis, greatly above the industry-reported average of 1–2%. More significantly, the exon-skipping candidate ENTR-601 for DMD offers the possibility of long-term muscle repair and regeneration by efficiently entering skeletal muscle progenitor cells in addition to delivering to mature muscle fibers [99]. PepGen employs linear peptides incorporating multiple non-natural amino acids. Its peptide-conjugated technology enhances endosomal escape efficiency of conjugated oligonucleotides by 94-fold compared to naked oligonucleotides and enables efficient targeted delivery to the nucleus, which is critical for diseases requiring action on nuclear RNA [100]. In the future, membrane-disrupting sequences derived from peptide library screening and rational design may be integrated into antibody scaffolds to construct next-generation smart carriers with highly efficient endosomal escape capabilities [101–105].

Third, a fundamental shift in therapeutic modalities and R&D paradigms is underway. In terms of payloads, the scope of AOCs is expanding from delivering gene regulatory tools like siRNA and ASO to delivering CRISPR-Cas gene editing systems. Vertex Pharmaceuticals' candidate drug VX-670 [106] couples an antibody targeting TfR1 with CRISPR components, aiming to achieve lifelong correction of genetic disorders with a single treatment. This marks AOC's transition from "reversible regulation" to a new phase of "permanent editing". In R&D methodologies, artificial intelligence (AI) is deeply integrating throughout the entire process—from target discovery, antibody/peptide design, affinity optimization, to pharmacokinetic prediction. AI-driven capabilities will significantly accelerate rational design processes, rapidly screening vast possibilities to identify optimal candidate molecules with high specificity, efficient endocytosis, and low immunogenicity. This approach systematically overcomes current R&D bottlenecks reliant on trial-and-error methodologies

[107, 108]. Dyne Therapeutics' FORCE™ technology platform has successfully achieved standardized production, shortening development cycles and reducing costs. This provides a scalable engineering foundation for rapid iteration and clinical translation of AOCs.

In summary, the AOC field is transitioning from isolated technological breakthroughs toward platform-based and intelligent development. Future success hinges on intelligently integrating diverse delivery strategies and cutting-edge therapeutic approaches like gene editing based on disease characteristics, while leveraging AI-driven rational design throughout the entire process. Through this deep cross-disciplinary integration, the platform holds promise for systematically overcoming extrahepatic delivery bottlenecks, ultimately delivering transformative precision therapies for numerous intractable diseases.

CONCLUSION

Over the past decade, oligonucleotide therapeutics have demonstrated significant growth in global clinical development. As a pivotal direction in this field, the research and development of AOCs is advancing rapidly. Although initially focused primarily on rare diseases, continuous technological innovation has progressively expanded the potential application scope of AOCs to broader therapeutic areas, showcasing the potential to transform treatment paradigms for chronic diseases. Compared to ADCs, AOCs face more distinctive challenges in molecular design, target selection, and tissue-specific delivery. Therefore, strengthening innovative exploration of targeting receptors and the strategic integration of novel technologies will be crucial to advancing the future development of AOCs. As design strategies continue to be optimized and technological platforms are further refined, AOCs are expected to emerge as a core technological platform in next-generation precision targeted therapies, following in the footsteps of ADCs, paving new paths for controllable, efficient, and durable gene regulation therapies.

REFERENCES

- Springer AD, Dowdy SF. GalNAc-siRNA conjugates: leading the way for delivery of RNAi therapeutics. *Nucleic Acid Ther.* 2018;28:109–118.
- Debacker AJ, Voutilainen J, Catley M, Blakey D, Habib N. Delivery of oligonucleotides to the liver with GalNAc: from research to registered therapeutic drug. *Mol Ther.* 2020;28:1759–1771.
- Wu X, Liu S, Liu D, Li X, Wang H, Han X. Global and Chinese trends in oligonucleotide drug clinical development: a comparative analysis. *Pharmacol Res.* 2024;210:107487.
- Zhang Y, Chen H, Hong L, Wang H, Li B, Zhang M, et al. Inclisiran: a new generation of lipid-lowering siRNA therapeutic. *Front Pharm.* 2023;14:1260921.
- Ray KK, Wright RS, Kallend D, Koenig W, Leiter LA, Raal FJ, et al. Two phase 3 trials of inclisiran in patients with elevated LDL cholesterol. *N Engl J Med.* 2020;382:1507–1519.
- Qin Z-X, Zuo L, Zeng Z, Ma R, Xie W, Zhu X, et al. GalNAc-siRNA conjugate delivery technology promotes the treatment of typical chronic liver diseases. *Expert Opin Drug Deliv.* 2025;22:455–469.
- Kaffe E, Roulis M, Zhao J, Qu R, Sefik E, Mirza H, et al. Humanized mouse liver reveals endothelial control of essential hepatic metabolic functions. *Cell.* 2023;186:3793–3809.e26.
- Malecova B, Burke RS, Cochran M, Hood MD, Johns R, Kovach PR, et al. Targeted tissue delivery of RNA therapeutics using antibody-oligonucleotide conjugates (AOCs). *Nucleic Acids Res.* 2023;51:5901–5910.
- Song E, Zhu P, Lee SK, Chowdhury D, Kussman S, Dykxhoorn DM, et al. Antibody mediated in vivo delivery of small interfering RNAs via cell-surface receptors. *Nat Biotechnol.* 2005;23:709–17.
- Cuellar TL, Barnes D, Nelson C, Tanguay J, Yu SF, Wen X, et al. Systematic evaluation of antibody-mediated siRNA delivery using an industrial platform of THIOMAB-siRNA conjugates. *Nucleic Acids Res.* 2015;43:1189–203.
- Janas MM, Schlegel MK, Harbison CE, Yilmaz VO, Jiang Y, Parmar R, et al. Selection of GalNAc-conjugated siRNAs with limited off-target-driven rat hepatotoxicity. *Nat Commun.* 2018;9:723.
- Zhang C, Dai P, Vinogradov AA, Gates ZP, Pentelute BL. Site-Selective Cysteine–Cyclooctyne Conjugation. *Angew Chem Int Ed.* 2018;57:6459–6463.
- Song X, Wang X, Ma Y, Liang Z, Yang Z, Cao H. Site-Specific Modification Using the 2'-Methoxyethyl Group Improves the Specificity and Activity of siRNAs. *Mol Ther - Nucleic Acids.* 2017;9:242–250.
- Geall AJ, Doppalapudi VR, Chu DS-H, Cochran MC, Hood M, Darimont BD, et al. Compositions and methods of treating muscle atrophy and myotonic dystrophy. *PubChem Patent Summary for US-10881743-B2.*
- Roberts TC, Langer R, Wood MJA. Advances in oligonucleotide drug delivery. *Nat Rev Drug Discov.* 2020;19:673–694.
- Meng Q, Yang M, Xing F, Xie Z, Hao Y, Jiang P, et al. Antibody-oligonucleotide conjugates in cancer therapy: potential and promise. *Crit Rev Oncol/Hematol.* 2025;215:104858.
- Su Z, Xiao D, Xie F, Liu L, Wang Y, Fan S, et al. Antibody-drug conjugates: Recent advances in linker chemistry. *Acta Pharm Sin B.* 2021;11:3889–3907.
- Rennick JJ, Johnston APR, Parton RG. Key principles and methods for studying the endocytosis of biological and nanoparticle therapeutics. *Nat Nanotechnol.* 2021;16:266–276.
- Candelaria PV, Leoh LS, Penichet ML, Daniels-Wells TR. Antibodies targeting the transferrin receptor 1 (TfR1) as direct anti-cancer agents. *Front Immunol.* 2021;12:607692.
- Aisen P. Transferrin receptor 1. *Int J Biochem Cell Biol.* 2004;36:2137–2143.
- Etzaniz U, Diaz M, Bhardwaj R, Tyaglo O, Lemoine K, Missinato MA, et al. Antibody-oligonucleotide conjugates (AOCs) demonstrate potent and durable exon skipping and dystrophin restoration in a mouse model of duchenne muscular dystrophy (S23.003). *Neurology.* 2022;98:1761.
- Girgenrath M, Estrella N, Kumar A, Li J, Hicks A, Brennan C, et al. P20 Endosomal Escape Vehicles (EEV™) - Oligonucleotides conjugates produce exon skipping and dystrophin production in preclinical models of Duchenne muscular dystrophy. *Neuromuscul Disord.* 2023;33:5102.
- Alam R, Alvarado A, Andrews FH, Babb NA, Balasubramaniam D, Chauvigne-Hines L, et al. Transferrin receptor binding proteins and conjugates. *Publication Number WO/2024/036096.*
- Lee HN, Jeong MS, Jang SB. Molecular characteristics of amyloid precursor protein (APP) and its effects in cancer. *Int J Mol Sci.* 2021;22:4999.
- Gordon RE, Nemeth JF, Singh S, Lingham RB, Grewal IS. Harnessing SLE auto-antibodies for intracellular delivery of biologic therapeutics. *Trends Biotechnol.* 2021;39:298–310.
- Rattray Z, Deng G, Zhang S, Shirali A, May CK, Chen X, et al. ENT2 facilitates brain endothelial cell penetration and blood-brain barrier transport by a tumor-targeting anti-DNA autoantibody. *JCI Insight.* 2021;6:e145875.
- Naes SM, Ab-Rahim S, Mazlan M, Abdul Rahman A, Zhang DM. Equilibrative nucleoside transporter 2: properties and physiological roles. *BioMed Res Int.* 2020;2020:5197626.
- Anderson CM, Baldwin SA, Young JD, Cass CE, Parkinson FE. Distribution of mRNA encoding a nitrobenzylthioinosine-insensitive nucleoside transporter (ENT2) in rat brain. *Brain Res Mol Brain Res.* 1999;70:293–7.
- Manso T, Sanou G, Nousias C, Maalem I, Boutin F, Giudicelli V, et al. Identification of engineered IMGT Fc variants in IMGT/mAb-DB, a database of therapeutic antibodies and fusion proteins. *MABs.* 2025;17:2594260.
- Rackear M, Quijano E, Ianniello Z, Colón-Ríos DA, Krysztofiak A, Abdullah R, et al. Next-generation cell-penetrating antibodies for tumor targeting and RAD51 inhibition. *Oncotarget.* 2024;15:699–713.
- Chen CF, Hsu EC, Lin KT, Tu PH, Chang HW, Lin CH, et al. Overlapping high-resolution copy number alterations in cancer genomes identified putative cancer genes in hepatocellular carcinoma. *Hepatology.* 2010;52:1690–701.
- Farré X, Guillén-Gómez E, Sánchez L, Hardisson D, Plaza Y, Lloberas J, et al. Expression of the nucleoside-derived drug transporters hCNT1, hENT1 and hENT2 in gynecologic tumors. *Int J Cancer.* 2004;112:959–66.
- Tufail M, Jiang C-H, Li N. Immune evasion in cancer: mechanisms and cutting-edge therapeutic approaches. *Signal Transduction Targeted Therapy.* 2025;10:227.
- Lanza F, Maffini E, Rondoni M, Massari E, Faini AC, Malavasi F. CD22 expression in B-cell acute lymphoblastic leukemia: biological significance and implications for inotuzumab therapy in adults. *Cancers.* 2020;12:303.
- Martino M, Alati C, Canale FA, Musuraca G, Martinelli G, Cerchione C. A review of clinical outcomes of CAR T-cell therapies for B-acute lymphoblastic leukemia. *Int J Mol Sci.* 2021;22:2150.
- Duan S, Paulson JC. Siglecs as immune cell checkpoints in disease. *Annu Rev Immunol.* 2020;38:365–395.
- Uy N, Nadeau Nguyen M, Stahl M, Zeidan A. Inotuzumab ozogamicin in the treatment of relapsed/refractory acute B cell lymphoblastic leukemia. *J Blood Med.* 2018;9:67–74.
- Kwong LS, Brown MH, Barclay AN, Hatherley D. Signal-regulatory protein α from the LOD mouse binds human CD47 with an exceptionally high affinity – implications for engraftment of human cells. *Immunology.* 2014;143:61–67.

39. Eladl E, Tremblay-LeMay R, Rastgoo N, Musani R, Chen W, Liu A, et al. Role of CD47 in hematological malignancies. *J Hematol Oncol*. 2020;13:96.
40. Yu J, Li S, Chen D, Liu D, Guo H, Yang C, et al. IMM0306, a fusion protein of CD20 mAb with the CD47 binding domain of SIRPα, exerts excellent cancer killing efficacy by activating both macrophages and NK cells via blockade of CD47-SIRPα interaction and FcγR engagement by simultaneously binding to CD47 and CD20 of B cells. *Leukemia*. 2023;37:695–698.
41. Weiskopf K, Jahchan NS, Schnorr PJ, Cristea S, Ring AM, Maute RL, et al. CD47-blocking immunotherapies stimulate macrophage-mediated destruction of small-cell lung cancer. *J Clin Invest*. 2016;126:2610–2620.
42. Roberts DD, Zhang M, Hutter G, Kahn SA, Azad TD, Gholamin S, et al. Anti-CD47 treatment stimulates phagocytosis of glioblastoma by M1 and M2 polarized macrophages and promotes M1 polarized macrophages in vivo. *Plos One*. 2016;11:e0153550.
43. Goulet DR, Atkins WM. Considerations for the design of antibody-based therapeutics. *J Pharm Sci*. 2020;109:74–103.
44. Patel DD, Bussell JB. Neonatal Fc receptor in human immunity: function and role in therapeutic intervention. *J Allergy Clin Immunol*. 2020;146:467–478.
45. Mackness BC, Jaworski JA, Boudanova E, Park A, Valente D, Mauriac C, et al. Antibody Fc engineering for enhanced neonatal Fc receptor binding and prolonged circulation half-life. *mAbs*. 2019;11:1276–1288.
46. Li Z, Krippendorff BF, Shah DK. Influence of Molecular size on the clearance of antibody fragments. *Pharm Res*. 2017;34:2131–2141.
47. Xu R, Zheng Y, Tai W. A single-chain fab derived drug conjugate for HER2 specific delivery. *Biomaterials*. 2025;313:122798.
48. Dumontet C, Reichert JM, Senter PD, Lambert JM, Beck A. Antibody–drug conjugates come of age in oncology. *Nat Rev Drug Discov*. 2023;22:641–661.
49. Richards DA. Exploring alternative antibody scaffolds: antibody fragments and antibody mimics for targeted drug delivery. *Drug Discov Today Technol*. 2018;30:35–46.
50. Smith BA, Meeks CB, Murray-Brown I, McCarthy EA, Julian MC, Arndt J, et al. Balancing brain exposure, pharmacokinetics and safety of transferrin receptor antibodies for delivery of neuro-therapeutics. *mAbs*. 2025;17:2592422.
51. Müller D. Targeting co-stimulatory receptors of the TNF superfamily for cancer immunotherapy. *BioDrugs*. 2023;37:21–33.
52. Yu X, Orr CM, Chan HTC, James S, Penfold CA, Kim J, et al. Reducing affinity as a strategy to boost immunomodulatory antibody agonism. *Nature*. 2023;614:539–547.
53. Segal NH, Logan TF, Hodi FS, McDermott D, Melero I, Hamid O, et al. Results from an integrated safety analysis of urelumab, an agonist anti-CD137 monoclonal antibody. *Clin Cancer Res*. 2017;23:1929–1936.
54. Foss S, Sakya SA, Aguinalde L, Lustig M, Shaughnessy J, Cruz AR, et al. Human IgG Fc-engineering for enhanced plasma half-life, mucosal distribution and killing of cancer cells and bacteria. *Nat Commun*. 2024;15:2007.
55. Østergaard ME, Carrer M, Anderson BA, Afetian M, Bakoosli Mohsen A, Santos JA, et al. Conjugation to a transferrin receptor 1-binding Bicycle peptide enhances ASO and siRNA potency in skeletal and cardiac muscles. *Nucleic Acids Res*. 2025;53:gkaf270.
56. Brennan FR, Polli JR, Sathish J, Ramones M, Wolf B, Schlothauer T, et al. Impact of antibody Fc engineering on translational pharmacology, and safety: insights from industry case studies. *mAbs*. 2025;17:2505092.
57. Dovgan I, Koniev O, Kolodych S, Wagner A. Antibody–oligonucleotide conjugates as therapeutic, imaging, and detection agents. *Bioconjugate Chem*. 2019;30:2483–2501.
58. Rattray Z, Dubljevic V, Rattray NJW, Greenwood DL, Johnson CH, Campbell JA, et al. Re-engineering and evaluation of anti-DNA autoantibody 3E10 for therapeutic applications. *Biochem Biophys Res Commun*. 2018;496:858–864.
59. Zhou L, Bi J, Chang S, Bai Z, Yu J, Wang R, et al. Self-Assembled Antibody-Oligonucleotide Conjugates for Targeted Delivery of Complementary Antisense Oligonucleotides. *Angew Chem Int Ed Engl*. 2025;64:e202415272.
60. van Geel R, Wijdeven MA, Heesbeen R, Verkade JMM, Wasie AA, van Berkel SS, et al. Chemoenzymatic conjugation of toxic payloads to the globally conserved N-glycan of native mAbs provides homogeneous and highly efficacious antibody–drug conjugates. *Bioconjugate Chem*. 2015;26:2233–2242.
61. Mullard A. Antibody-oligonucleotide conjugates enter the clinic. *Nat Rev Drug Discov*. 2022;21:6–8.
62. Bargh JD, Isidro-Llobet A, Parker JS, Spring DR. Cleavable linkers in antibody-drug conjugates. *Chem Soc Rev*. 2019;48:4361–4374.
63. Subramanian RR, Qatanani MT, Weeden T, Desjardins CA. Muscle targeting complexes and uses thereof for treating myotonic dystrophy. *PubChem Patent Summary for US-2024148891-A1*.
64. Desjardins CA, Tang K, Beskrovnyaya O, Davis J, Mix C, Weeden T et al. Dosing Of Muscle Targeting Complexes For Treating Dystrophinopathies. *PubChem Patent Summary for US-2025018050-A1*.
65. Subramanian RR, Qatanani MT, Weeden T, Desjardins CA, Quinn B, Najim J. Muscle targeting complexes and uses thereof for skipping exon 45 OF A DMD GENE. *Publication Number WO/2024/097644*.
66. Su D, Kozak KR, Sadowsky J, Yu S-F, Fourie-O'Donohue A, Nelson C, et al. Modulating antibody–drug conjugate payload metabolism by conjugation site and linker modification. *Bioconjugate Chem*. 2018;29:1155–1167.
67. Hofer K, Mansø M, Martin RE, Mohamed MYH, Oelschlaegel T, Ravn J, et al. Linkers for site-specific antibody conjugation. *Publication Number WO 2024/017923 A1*.
68. Cochran M, Arias D, Burke R, Chu D, Erdogan G, Hood M, et al. Structure-activity relationship of antibody-oligonucleotide conjugates: evaluating bioconjugation strategies for antibody-siRNA conjugates for drug development. *J Med Chem*. 2024;67:14852–14867.
69. Nair JK, Attarwala H, Sehgal A, Wang Q, Aluri K, Zhang X, et al. Impact of enhanced metabolic stability on pharmacokinetics and pharmacodynamics of GalNAc-siRNA conjugates. *Nucleic Acids Res*. 2017;45:10969–10977.
70. Brown CR, Gupta S, Qin J, Racie T, He G, Lentini S, et al. Investigating the pharmacodynamic durability of GalNAc-siRNA conjugates. *Nucleic Acids Res*. 2020;48:11827–11844.
71. Schlegel MK, Foster DJ, Kel'in AV, Zlatev I, Bisbe A, Jayaraman M, et al. Chirality dependent potency enhancement and structural impact of glycol nucleic acid modification on siRNA. *J Am Chem Soc*. 2017;139:8537–8546.
72. Li P, Zhang J, Jia Y, Zhang Y. Novel mono-PEGylated dimeric GLP-1 conjugate with enhanced receptor activation and prolonged anti-diabetes efficacies. *Life Sci*. 2020;254:117752.
73. Bennett CF. Therapeutic antisense oligonucleotides are coming of age. *Annu Rev Med*. 2019;70:307–321.
74. Geary RS, Norris D, Yu R, Bennett CF. Pharmacokinetics, biodistribution and cell uptake of antisense oligonucleotides. *Adv Drug Deliv Rev*. 2015;87:46–51.
75. Wang J, Tan M, Wang Y, Liu X, Lin A. Advances in modification and delivery of nucleic acid drugs. *Zhejiang Da Xue Xue Bao Yi Xue Ban*. 2023;52:417–428.
76. Hu B, Zhong L, Weng Y, Peng L, Huang Y, Zhao Y, et al. Therapeutic siRNA: state of the art. *Signal Transduct Target Ther*. 2020;5:101.
77. Kozma GT, Shimizu T, Ishida T, Szebeni J. Anti-PEG antibodies: Properties, formation, testing and role in adverse immune reactions to PEGylated nanobiopharmaceuticals. *Adv Drug Deliv Rev*. 2020;154:155:163–175.
78. Liu M, Li J, Zhao D, Yan N, Zhang H, Liu M, et al. Branched PEG-modification: A new strategy for nanocarriers to evade of the accelerated blood clearance phenomenon and enhance anti-tumor efficacy. *Biomaterials*. 2022;283:121415.
79. Wang Z, Wang X, Zhou L, Shi S, Hua Y, Feng Y. Safety and efficacy of 48-week pegylated interferon-α-2b therapy in patients with hepatitis B virus-related compensated liver cirrhosis: a pilot observational study. *Front Med*. 2024;11:1489671.
80. Guan G, Zhang T, Ning J, Tao C, Gao N, Zeng Z, et al. Higher TP53BP2 expression is associated with HBsAg loss in peginterferon-α-treated patients with chronic hepatitis B. *J Hepatol*. 2024;80:41–52.
81. Hagiwara K, Kurihara K, Honma M, Yamamoto J, Shinohara F. PEG-modification on the endo-position of an antisense oligonucleotide increases tumor accumulation via the EPR effect. *J Biomater Sci, Polym Ed*. 2018;29:448–459.
82. Hawner M, Ducho C. Cellular targeting of oligonucleotides by conjugation with small molecules. *Molecules*. 2020;25:5963.
83. Hörberg J, Carlesso A, Reymer A. Mechanistic insights into ASO-RNA complexation: advancing antisense oligonucleotide design strategies. *Mol Therapy - Nucleic Acids*. 2024;35:102351.
84. Ramasamy T, Ruttala HB, Munusamy S, Chakraborty N, Kim JO. Nano drug delivery systems for antisense oligonucleotides (ASO) therapeutics. *J Controlled Release*. 2022;352:861–878.
85. Nan Y, Zhang Y-J. Antisense phosphorodiamidate morpholino oligomers as novel antiviral compounds. *Front Microbiol*. 2018;9:750.
86. Eteplirsen. In: *LiverTox: Clinical and Research Information on Drug-Induced Liver Injury*. National Institute of Diabetes and Digestive and Kidney Diseases: Bethesda (MD), 2012.
87. Heo YA. Golodirsen: first approval. *Drugs*. 2020;80:329–333.
88. Clemens PR, Rao VK, Connolly AM, Harper AD, Mah JK, Smith EC, et al. Safety, tolerability, and efficacy of viltolarsen in boys with duchenne muscular dystrophy amenable to exon 53 skipping: a phase 2 randomized clinical trial. *JAMA Neurol*. 2020;77:982–991.
89. Shirley M. Casimersen: first approval. *Drugs*. 2021;81:875–879.
90. Desjardins CA, Yao M, Hall J, O'Donnell E, Venkatesan R, Spring S, et al. Enhanced exon skipping and prolonged dystrophin restoration achieved by TfR1-targeted delivery of antisense oligonucleotide using FORCE conjugation in mdx mice. *Nucleic Acids Res*. 2022;50:11401–11414.
91. Etxaniz U, Marks I, Albin T, Diaz M, Bhardwaj R, Anderson A, et al. AOC 1044 induces exon 44 skipping and restores dystrophin protein in preclinical models of Duchenne muscular dystrophy. *Nucleic Acids Res*. 2025;53:gkaf241.

92. Huo S, Li H, Boersma AJ, Herrmann A. DNA nanotechnology enters cell membranes. *Adv Sci*. 2019;6:1900043.
93. Wan HI, Bermingham CNWL. Methods of treating cancer using anti-Cd22 antibody oligonucleotide conjugates. PubChem Patent Summary for WO-2023225577-A1.
94. Walker I, Irwin WJ, Akhtar S. Improved cellular delivery of antisense oligonucleotides using transferrin receptor antibody-oligonucleotide conjugates. *Pharm Res*. 1995;12:1548–53.
95. Lehot V, Kuhn I, Nothisen M, Erb S, Kolodych S, Cianféroni S, et al. Non-specific interactions of antibody-oligonucleotide conjugates with living cells. *Sci Rep*. 2021;11:5881.
96. Heh E, Allen J, Ramirez F, Lovasz D, Fernandez L, Hogg T, et al. Peptide drug conjugates and their role in cancer therapy. *Int J Mol Sci*. 2023;24:829.
97. Peters C, Brown S. Antibody-drug conjugates as novel anti-cancer chemotherapeutics. *Biosci Rep*. 2015;35:e00225.
98. Lee HJ, Xie Y, Greineder CF, Tessier PM Bioconjugates for improved delivery of oligonucleotide therapeutics to the central nervous system. *Adv Drug Delivery Rev*. 2026;230:115778.
99. Oldham M, Estrella N, Kumar A, Hicks A, Brennan C, Blake S, et al. 433P Therapeutic potential of ENTR-601-44, an Endosomal Escape Vehicle (EEV™) - Oligonucleotide Conjugate for the treatment of exon 44 skip amenable DMD. *Neuromuscul Disord*. 2024;43:104441.304.
100. Larkindale J, Shoskes J, Garg B, Song G, Lonkar P, Babcock S, et al. 461P Evaluation of PGN-EDODM1: FREEDOM-DM1 and FREEDOM2-DM1 clinical trials in myotonic dystrophy type 1. *Neuromuscul Disord*. 2024;43:104441.536.
101. Hammond SM, Aartsma-Rus A, Alves S, Borgos SE, Buijsen RAM, Collin RWJ, et al. Delivery of oligonucleotide-based therapeutics: challenges and opportunities. *EMBO Mol Med*. 2021;13:e13243.
102. Ämmälä C, Drury WJ, Knerr L, Ahlstedt I, Stillemark-Bilton P, Wennberg-Huldt C, et al. Targeted delivery of antisense oligonucleotides to pancreatic β -cells. *Sci Adv*. 2018;4:eaat3386.
103. Knerr L, Prakash TP, Lee R, Drury Iii WJ, Nikan M, Fu W, et al. Glucagon Like Peptide 1 Receptor Agonists for Targeted Delivery of Antisense Oligonucleotides to Pancreatic Beta Cell. *J Am Chem Soc*. 2021;143:3416–3429.
104. Kulkarni JA, Witzigmann D, Thomson SB, Chen S, Leavitt BR, Cullis PR, et al. The current landscape of nucleic acid therapeutics. *Nat Nanotechnol*. 2021;16:630–643.
105. Klabenkova K, Fokina A, Stetsenko D. Chemistry of peptide-oligonucleotide conjugates: a review. *Molecules*. 2021;26:5420.
106. Paker A, Lochmüller H, Turner C, Nigim F, Larimer P, Johnson N. 386PA phase 1/2 trial (Galileo Study) of VX-670 in adults with myotonic dystrophy type 1 (DM1): study design. *Neuromuscul Disord*. 2025;53:105952.
107. Zhang K, Yang X, Wang Y, Yu Y, Huang N, Li G, et al. Artificial intelligence in drug development. *Nat Med*. 2025;31:45–59.
108. Sun Q, Wang H, Xie J, Wang L, Mu J, Li J, et al. Computer-aided drug discovery for undruggable targets. *Chem Rev*. 2025;125:6309–6365.
109. Yu J, Song Y, Tian W. How to select IgG subclasses in developing anti-tumor therapeutic antibodies. *J Hematol Oncol*. 2020;13:45.
110. De Cecco M, Galbraith DN, McDermott LL. What makes a good antibody-drug conjugate? *Expert Opin Biol Ther*. 2021;21:841–847.
111. Li M, An H, Zhang J, Li W, Yu C, Wang L. Advances in the pharmaceutical development of antibody-oligonucleotide conjugates. *Eur J Pharma Sci*. 2025;215:107292.
112. Jain N, Smith SW, Ghone S, Tomczuk B. Current ADC linker chemistry. *Pharm Res*. 2015;32:3526–40.
113. Tsuchikama K, An Z. Antibody-drug conjugates: recent advances in conjugation and linker chemistries. *Protein Cell*. 2018;9:33–46.
114. Okamura K, Ishizuka A, Siomi H, Siomi MC. Distinct roles for Argonaute proteins in small RNA-directed RNA cleavage pathways. *Genes Dev*. 2004;18:1655–66.
115. Shah MNA, Wilton-Clark H, Haque F, Powell B, Sutanto LE, Maradiya R, et al. DG9 boosts PMO nuclear uptake and exon skipping to restore dystrophic muscle and cardiac function. *Nat Commun*. 2025;16:4477.
116. Zhang Z, Kuo JC, Yao S, Zhang C, Khan H, Lee RJ. CpG oligodeoxynucleotides for anticancer monotherapy from preclinical stages to clinical trials. *Pharmaceutics*. 2021;14:73.
117. Li M, Yao H, Yi K, Lao YH, Shao D, Tao Y. Emerging nanoparticle platforms for CpG oligonucleotide delivery. *Biomater Sci*. 2024;12:2203–2228.
118. Qi QR, Tian H, Yue BS, Zhai BT, Zhao F. Research progress of SN38 drug delivery system in cancer treatment. *Int J Nanomed*. 2024;19:945–964.
119. Norsworthy KJ, Ko CW, Lee JE, Liu J, John CS, Przepiorka D, et al. FDA approval summary: mylotarg for treatment of patients with relapsed or refractory CD33-positive acute myeloid leukemia. *Oncologist*. 2018;23:1103–1108.
120. Juliano RL. The delivery of therapeutic oligonucleotides. *Nucleic Acids Res*. 2016;44:6518–48.
121. Joubert N, Beck A, Dumontet C, Denevault-Sabourin C. Antibody-drug conjugates: the last decade. *Pharmaceutics (Basel)*. 2020;13:245.
122. Sheridan C. Now with oligos: antibody-oligonucleotide conjugates are the new drug modality to watch. *Nat Biotechnol*. 2026;44:3–5.
123. Beck A, Goetsch L, Dumontet C, Corvaia N. Strategies and challenges for the next generation of antibody-drug conjugates. *Nat Rev Drug Discov*. 2017;16:315–337.
124. Fu Z, Li S, Han S, Shi C, Zhang Y. Antibody drug conjugate: the “biological missile” for targeted cancer therapy. *Signal Transduct Target Ther*. 2022;7:93.
125. Brown DG, Wobst HJ. A decade of FDA-approved drugs (2010–2019): trends and future directions. *J Med Chem*. 2021;64:2312–2338.
126. Maecker H, Jonnalagadda V, Bhakta S, Jammalamadaka V, Junutula JR. Exploration of the antibody-drug conjugate clinical landscape. *mAbs*. 2023;15:2229101.
127. Noble PW, Bernatsky S, Clarke AE, Isenberg DA, Ramsey-Goldman R, Hansen JE. DNA-damaging autoantibodies and cancer: the lupus butterfly theory. *Nat Rev Rheumatol*. 2016;12:429–434.
128. Veillette A, Chen J. SIRPα-CD47 immune checkpoint blockade in anticancer therapy. *Trends Immunol*. 2018;39:173–184.
129. Paunovska K, Loughrey D, Dahlman JE. Drug delivery systems for RNA therapeutics. *Nat Rev Genet*. 2022;23:265–280.
130. Malecova B, Burke R, Darimont BD, SALA Cano D. Compositions and methods of treating facioscapulohumeral muscular dystrophy. PubChem Patent Summary for US-2023279395-A1.
131. Ludwig DL, Pamarthy PBS, Ahmed RA, Bommireddy Venkata V, Quijano E, Glazer P. Antibody oligonucleotide conjugates. Publication Number WO/2024/158824.

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AUTHOR CONTRIBUTIONS

Wsf generated the main idea, performed literature search and analysis, and discussed and aligned all literature data and wrote the manuscript. WJL and WJY helped write the manuscript and design the figures. CW and Xfl prepared the tables. YIF, HqC, and CyF conceived the manuscript. YL and LN discussed and reviewed the manuscript. All authors reviewed and approved the final manuscript.

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COMPETING INTERESTS

The authors declare no competing interests.

ADDITIONAL INFORMATION

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